Hugging Face, Model Selection & Unsloth

Advanced Data / AI — Learning Review

Day

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Objectives

- Discover pretrained models on Hugging Face
- Select the right architecture for a task
- Load models with `Transformers`
- Understand how Unsloth makes fine-tuning efficient

Hugging Face Ecosystem

Component	Purpose
Models	Discover pretrained models
Datasets	Training and evaluation data
Spaces	Deploy interactive AI apps
Buckets	Store files and model assets
Docs	Libraries and API guides
Enterprise	Enterprise AI solutions

Key Concepts

Pretrained model	Already learned patterns from data
Tokenizer	Converts text to tokens
Architecture	Structure of the neural network
Weights	Learned parameters
Inference	Using a trained model to predict
Fine-tuning	Further training on a task-specific dataset

Loading Models: `from_pretrained()`

```python
from transformers import AutoTokenizer, AutoModel

model_name = "bert-base-uncased"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModel.from_pretrained(model_name)
```

1. Identify model $\rightarrow$ load config & architecture
2. Retrieve pretrained weights
3. Load tokenizer
4. Ready for inference or further training

Loading $\neq$ fine-tuning.

AutoModel Variants

Loader	Purpose	Example
AutoModel	General representation	Embeddings
...ForSequenceClassification	Classify sequences	Sentiment
...ForTokenClassification	Classify tokens	NER
...ForQuestionAnswering	Extract answers	Reading comprehension
...ForCausalLM	Autoregressive generation	GPT-style text
...ForMaskedLM	Predict masked tokens	BERT-style
...ForSeq2SeqLM	Sequence $\rightarrow$ sequence	Translation, summary
...ForSpeechSeq2Seq	Speech-to-text	ASR
...ForImageSegmentation	Pixel-level analysis	Segmentation

Causal LM vs. Masked LM

Causal LM

Predicts the next token from previous tokens.

Financial markets are $\rightarrow$ **unpredictable**

Masked LM

Predicts hidden tokens from surrounding context.

Financial [MASK] are unpredictable $\rightarrow$ **markets**

Model Selection: 5 Factors

1. **Task alignment** — match pretraining objective
2. **Architecture** — suitability and modularity
3. **Size & complexity** — balance with compute
4. **Benchmarking** — existing performance, transferability
5. **Community & docs** — support and resources

The best model is not the largest — it fits the task within the resources.

What is Unsloth?

- Framework for **efficient LLM fine-tuning**
- Lower GPU memory usage, faster training
- Optimized model loading
- LoRA / QLoRA support
- Integrates with Hugging Face; saves and exports models

Unsloth is a tool, not a pretrained model.

LoRA vs. QLoRA

LoRA

Trains small low-rank matrices instead of all parameters.

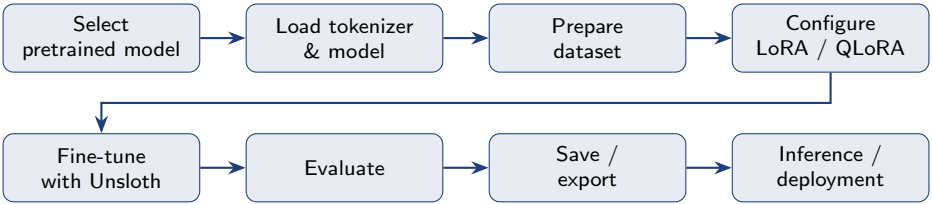
- Fewer trainable parameters
- Lower memory, faster
- Small adapter files

QLoRA

Quantized base model + LoRA adapters.

- Much lower GPU memory
- Larger models become accessible
- Base weights stay frozen

Workflow: Hugging Face + Unsloth



Unsloth Loading Example

```
from unsloth import FastLanguageModel

model, tokenizer = FastLanguageModel.from_pretrained(
    model_name="your-model-name",
    max_seq_length=2048,
    load_in_4bit=True
)
```

Model names, hardware needs, and training configs depend on the chosen model and environment.

Hugging Face vs. Unsloth

Aspect	Hugging Face	Unsloth
Main role	Model ecosystem & libraries	Optimized fine-tuning
Model discovery	Yes	Not primary
Pretrained models	Provides access	Loads supported models
Fine-tuning	Supported	Optimized workflows
LoRA / QLoRA	Via compatible libs	Integrated

Complementary, not replacements.

Fair Benchmarking

Control

- Same pretrained model
- Same dataset and split
- Same GPU / hardware
- Same training config

Measure

- Training time
- GPU memory
- Training / evaluation loss
- Task-specific metrics

Speed alone does not prove a better model.

Takeaways

- Hugging Face = ecosystem; Unsloth = efficient fine-tuning
- Pick the loader that matches the task head
- Choose models by fit, not size
- LoRA / QLoRA cut memory and cost
- Benchmark under controlled conditions

References

- Hugging Face Model Hub
- Hugging Face Transformers Documentation
- Hugging Face Datasets
- Unsloth Documentation